

SCIENCE HIERARCHOGRAPHY: Hierarchical Organization of Science Literature

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Abstract

Scientific knowledge is growing rapidly, making it challenging to track progress and high-level conceptual links across broad disciplines. While existing tools like citation networks and search engines make it easy to access a few related papers, they fundamentally lack the flexible abstraction needed to represent the *density* of activity in various scientific subfields.

We motivate SCIENCE HIERARCHOGRAPHY, the goal of organizing scientific literature into a high-quality hierarchical structure that allows for the categorization of scientific work across varying levels of abstraction, from very broad fields to very specific studies. Such a representation can provide insights into which fields are well-explored and which are under-explored. To achieve the goals of SCIENCE HIERARCHOGRAPHY, we develop a range of algorithms. Our primary approach combines fast embedding-based clustering with LLM-based prompting to balance the computational efficiency of embedding methods with the semantic precision offered by LLM prompting. We demonstrate that this approach offers the best trade-off between quality and speed compared to methods that heavily rely on LLM prompting, such as iterative tree construction with LLMs. To better reflect the interdisciplinary and multifaceted nature of research papers, our hierarchy captures multiple dimensions of categorization beyond simple topic labels. We evaluate the utility of our framework by assessing how effectively an LLM-based agent can locate target papers using the hierarchy. Results show that this structured approach enhances interpretability, supports trend discovery, and offers an alternative pathway for exploring scientific literature beyond traditional search methods.¹

1 Introduction

The publication of scholarly work has been increasing significantly in recent years (Ware & Mabe, 2015). Despite the overall growth, this expansion is not uniformly distributed across all scientific fields. In fact, certain areas likely receive more attention than others, presenting both opportunities and challenges (Hope et al., 2023). Therefore, a natural question is:

do we have a mechanism to understand the distribution of scientific efforts across different sub-areas?

Answering this question is crucial, as academic institutions can identify emerging or underexplored research areas, and policymakers can make more informed decisions about funding allocation based on the actual distribution of scholarly work, potentially directing more resources to important but under-explored areas.

Among existing solutions, conventional tools for discovering scientific literature, search engines like Google Scholar primarily function as retrieval engines. They are designed to find the few, most relevant literature matching a specific query rather than providing a

¹Code, data and demo: <https://github.com/JHU-CLSP/science-hierarchy>

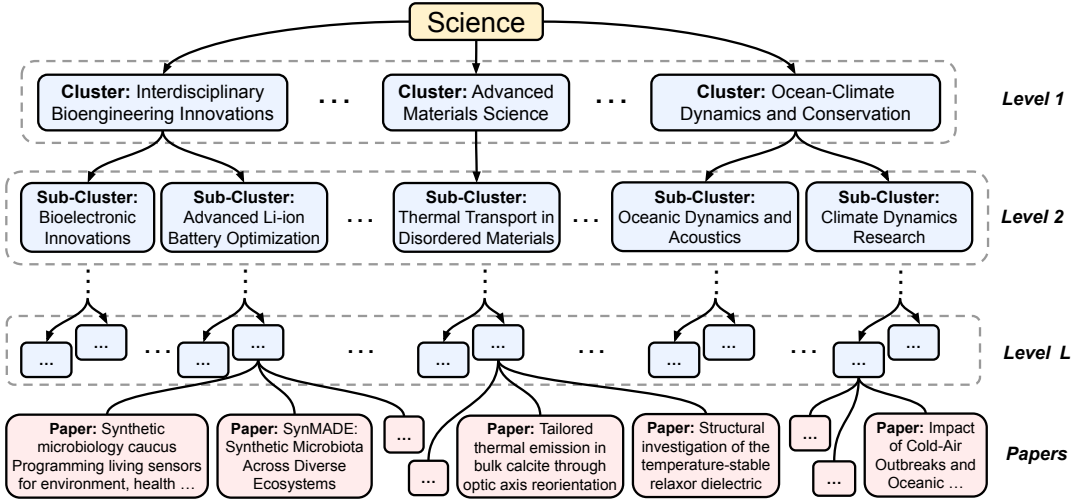


Figure 1: An example of SCIENCE HIERARCHOGRAPHY which illustrates the organization of scholarly work. Starting from broad research fields at the top levels, narrowing down through sub-clusters, and ultimately reaching specific papers within a particular research domain at the bottom level. Crucially, this must be inferred at scale and automatically.

comprehensive, structured overview of the entire scientific landscape. Similarly, modern LLM-based chatbots can recite related works if they were observed during pre-training, but again it’s not easy to glean a bird’s-eye view of scientific progress.

Understanding the distribution properties of science fundamentally requires a mechanism to abstract over scientific concepts (such as open science problems and techniques) and connect these abstractions to various, more concrete instances. At one end of the spectrum, we have broad scientific communities (like physics or artificial intelligence), and at the most specific level, we have papers published in these areas. Between these two endpoints lie intermediate concepts that group together sub-areas. Essentially, we need a data structure that encompasses both these endpoints of abstraction and various in-between abstractions.

We propose the construction of large-scale hierarchical representations of scientific literature, which we refer to as SCIENCE HIERARCHOGRAPHY. A well-designed hierarchy offers several benefits: it provides a macro-level view of scientific progress, showing how research efforts are distributed across different methodologies and application areas. This insight helps researchers identify emerging trends and gaps, while aiding policymakers in more strategic resource allocation. Additionally, such a hierarchy offers an alternative way to explore literature, complementing traditional retrieval systems by enabling users to navigate science through conceptual hierarchies rather than isolated keyword searches.

How should one represent scholarly works? A key challenge in building a hierarchical structure for scientific literature is defining the unit of representation for individual nodes. Research papers often span multiple topics (for example, *reinforcement learning applied to medical imaging*” or *deep learning for oceanography*”). We devise a prompting strategy to break down papers along multiple key *contribution* types (such as the problems they tackle and the techniques they address), capturing their multifaceted nature. For a fixed contribution type from science papers, we construct their hierarchical representation. Our approach ensures that papers are structured under meaningful and coherent categories (§3.2).

What are construction strategies that allow for both scalability and quality? To tackle this goal, we propose SCYCHIC (pronounced “psychic”), a novel approach to constructing a high-quality hierarchical structure for organizing scientific literature. SCYCHIC combines fast embedding-based clustering with LLM-based prompting to balance the computational efficiency of embedding methods with the semantic precision of LLM prompting (§4.1).

How can we assess the quality of a given science hierarchy? There is no definitive ground truth for a scientific hierarchy, as it naturally evolves over time—today’s scientific hierarchy differs from that of last year. We approach our **evaluation through utilization**,

focusing on whether an information seeker (human or AI agent) can efficiently locate specific information (such as details related to child nodes) by navigating the hierarchy from its root. Our evaluation is based on the premise that an effective hierarchy facilitates rapid information discovery, even though its applications extend beyond mere discovery (§5.2).

What did we empirically observe? We demonstrate that this approach offers the best trade-off between quality and speed compared to methods that heavily rely on LLM prompting, such as iterative tree construction or iterative pruning algorithms with LLMs. Through extensive experiments, we show that SCYCHIC, our method, can construct higher-quality hierarchies compared to a wide range of baseline approaches (§5.4). Meanwhile, validation of our approach on a 10K paper dataset demonstrates robust accuracy, confirming its scalability for large-scale applications.

Contributions: (1) We propose the goal of constructing large-scale abstract hierarchies for scientific literature, which can provide essential insights into the distribution of scholarly work across various fields. (2) We establish an evaluation framework that frames it as utilization—effective information discovery by traversal of the hierarchy. (3) We present SCYCHIC, a novel approach that integrates fast embedding-based clustering with LLM-based prompting to develop a high-quality hierarchical structure, tailored to capture different dimensions of academic work. Our extensive experiments demonstrate that our approach constructs superior hierarchies compared to existing baseline methods, ultimately paving the way for a more organized and insightful understanding of scientific progress.

2 Related Work

Approaches to hierarchy induction: Hierarchy (taxonomy) induction has evolved significantly from early pattern-based methods to sophisticated, LLM-augmented frameworks. The pioneering work of Hearst (1992) initiated the field by leveraging hand-crafted hyponym patterns to bootstrap is-a relation extraction. Subsequent studies advanced these ideas by combining statistical techniques with large-scale information extraction to automatically identify hypernym-hyponym patterns (Pantel & Pennacchiotti, 2006; Yang & Callan, 2009; Girju et al., 2006). More recent efforts have incorporated LLMs and iterative prompting to refine and complete taxonomies; for example, Wan et al. (2024); Zeng et al. (2024a); Chen et al. (2023); Zeng et al. (2024b) utilized zero-shot/few-shot reasoning and ensemble-based ranking for hierarchy construction. Complementary approaches include autonomous creation of fine-grained taxonomies without fixed vocabularies (Gunn et al., 2024), self-supervised expansion in low-resource domains (Mishra et al., 2024), and graph-based methods that integrate metadata and citation information (Cong et al., 2024; Sas & Capiluppi, 2024; Shen et al., 2024). Additional works further demonstrate the effectiveness of optimization and in-context learning (Hu et al., 2024b; Shi et al., 2024; Xu et al., 2025; Jain & Espinosa Anke, 2022; Chen et al., 2021). Our work differs from existing literature in problem scope, scale, and algorithmic solutions. Our aim is to progress toward scaling taxonomy induction for the domain of scholarly works, which is more challenging than prior domains in taxonomy induction (e.g., entity names) due to the complexity of scientific works, their size and evolving nature. The closest work to ours is by Oarga et al. (2024), but our broader scope requires different algorithmic approaches and lacks ground truth labels.

Clustering with LLMs: Recent advances in clustering methodologies augmented by LLMs have demonstrated effective ways to generate interpretable groupings of text. For example, Viswanathan et al. (2024); Katz et al. (2024) apply few-shot clustering and thematic grouping to partition scientific literature into meaningful subtopics, while Zhang et al. (2023); Wang et al. (2023) further refine these techniques by aligning clustering outcomes with natural language explanations and user intent. Other recent work iteratively refines cluster representations by replacing cluster centroids or summary points with LLM-generated natural language descriptions and inclusion criteria, thereby inducing more abstract, interpretable concepts over multiple clustering rounds (Lam et al., 2024; Diaz-Rodriguez, 2025). While these approaches improve clustering quality by using LLMs at various stages, they mostly result in flat groupings rather than hierarchical structures. Our approach builds on this by using LLMs to cluster documents and organizing these clusters into a structured hierarchy.

Structured representation of science: Scientific knowledge is expanding at an unprecedented pace (Teufel et al., 1999; Pertsas & Constantopoulos, 2017; Constantin et al., 2016; Fisas et al., 2016; Liakata et al., 2010), prompting the development of various frameworks for constructing structured representations and knowledge graphs (Fathalla et al., 2017; Jaradeh et al., 2019; Oelen et al., 2020; Vogt et al., 2020; Soldatova & King, 2006). For instance, Pham et al. (2024) develop a prompt-based method for generating interpretable topics with natural language labels, while Cui et al. (2024) and Marchenko & Dvoichenkov (2024) propose iterative frameworks that integrate object properties and graph mining techniques to refine taxonomies. Similarly, Zimmermann et al. (2024) combine curated ontologies with data-driven topic maps to offer an intuitive, multi-topic navigation system. Building on and extending these works, our work focuses on constructing a high-quality hierarchical structure for the scientific literature domain.

3 SCIENCE HIERARCHOGRAPHY: Toward Hierarchy of Scholarly Work

We begin with a formal definition of the problem (§3.1), introduce our evaluation (§5.2), and discuss the design choices on representing scientific content in our hierarchy (Section 3.2).

3.1 Formal Problem Statement

We define the task of SCIENCE HIERARCHOGRAPHY as an inference problem where the **input** is a large set of scientific papers: $P = \{p_1, p_2, \dots, p_n\}$. The goal is to infer a hierarchical structure for these papers (e.g., their problems and techniques). This structure represents levels of specificity and abstraction, with nodes closer to the root representing broader topics. Broader topics are at the upper levels, while more specific subtopics and individual papers are at the lower levels.

Formally, this goal is defined as inferring a graph that encodes relationships between nodes (atomic concepts representing scholarly ideas or goals). This can be formally defined as a tuple $T := (N, E)$ where:

- N is the set of all concepts (nodes) present in the hierarchy: $N = \{n_1, n_2, \dots, n_N\}$.
- E is the set of all edges connecting node pairs: $E = \{r_1, r_2, \dots, r_M\}$ where each edge is a node pair $r_i = (n_p, n_c)$ connects a parent node n_p to its child node n_c .
- For each edge $e \in E$, there is a type attribute: $\text{type}(e) \in \{\text{"isA"}, \text{"instanceOf"}\}$ associated with the edge.
- Each node $n \in N$ has a property $n.\text{prop}$ which defines the content of the node (e.g., a sentence description or a few keywords).

The edge (relations connecting two nodes) have types that is either "isA" relationships or "instanceOf." The "isA" relationship defines a hierarchical link between node pairs, indicating a child node is a subclass of its more abstract parent node (e.g., "RLHF isA RL" means "RLHF" is a type of "RL"). The "instanceOf" relationship indicates that a specific instance belongs to a concept category (e.g., "Ouyang et al. (2022) instanceOf RLHF" means "Ouyang et al. (2022)" is a specific instance of the class "RLHF").

3.2 Decomposing Each Science Paper into Atomic Contributions

A key question is how to represent scholarly work content within hierarchy nodes. Science documents are rich in ideas, covering broad goals, specific problems, and techniques. This necessitates extracting structured representations of papers (D’Souza & Auer, 2020) to disentangle their different focuses. This also addresses length challenges: Most papers are 4 to 10 pages, spanning 5K to 10K tokens. Processing all this information for many papers would be prohibitively challenging and costly for any language model.

We use an LLM to preprocess each paper (title and abstract) and break them down into a **pre-defined set of contributions**. We consider the following types: **(1) problem statement** (the problem addressed), **(2) solution** (the technical approach used), **(3) result** (the key

finding), and **(4) topic** (the overarching themes). Prompts for extracting contributions and examples are detailed in §C. We note that each contribution may include additional dimensions (sub-contributions). For instance, a “result” encompasses both the “outcome” and its “potential impact.”

3.3 Choosing the Effective Number of Layers in the Hierarchy

Although the effective number of layers is an empirical question, we can derive useful intuition by considering the structure of a near-balanced tree. Suppose the tree has L layers and a branching factor of b . Such a tree will contain approximately $O(b^L)$ nodes. If we aim to organize C contributions into a near-balanced hierarchy, we must ensure that the total number of leaf and intermediate nodes grows proportionally with C . To achieve this, the required depth L must scale logarithmically with the number of contributions, specifically as $L = O(\log_b C)$. In our experiments, we selected $L = 3$ for a corpus of 2K papers and $L = 4$ for 10K papers, both aligning with this logarithmic relationship. Extrapolating from this trend, if we were to scale up the corpus size to a range of 10^6 to 10^7 papers, we would expect the necessary depth to increase to approximately $L = 6$ or $L = 7$.

4 A Spectrum of Approaches to Tackle SCIENCE HIERARCHOGRAPHY

We present algorithms to address our proposed goal. We begin with our approach (SCYCHIC; §4.1), its special cases (§4.2) and then discuss other baselines that heavily rely on LLMs (FLMSCI; §4.3). While all these methods utilize LLMs, they vary in their reliance on them; some require many LLM calls (linear or quadratic relative to the number of papers), while others require fewer (logarithmic relative to the number of papers). Given the aim to scale to millions of papers, scalability is crucial. The objective is to find the approach that produces the highest quality hierarchy with the fewest LLM calls, which are costly in terms of latency and finances.

4.1 SCYCHIC: Alternating Between Clustering and Summarization

This approach is based on the following design choices: (1) access to **embedder**, a neural model that converts a description into a d -dimensional vector, (ideally) capturing its semantic meaning; (2) a clustering algorithm **clusterer** that, given the hyperparameter k , generates k clusters; (3) a contribution type (e.g., problem definition) and its dimensions C extracted per paper as detailed in §3.2 which determines the focus of the node descriptions; (4) **summarizer**, an LLM that generates a summary description which (ideally) provides a more abstract description of a collection of node descriptions; and (5) the total number of hierarchy layers L and target number of clusters in each layer $(k_1, k_2, \dots, k_L)$.

Initialization: The approach begins by embedding each papers. For each paper p_i , we embed each component in C : $\text{embedder}(c_j^i) \in \mathbb{R}^d$, where $j \in C$. This process results in $|C|$ embeddings per paper. We concatenate these embeddings, yielding $\mathbb{R}^{d \cdot |C|}$ embeddings per paper. With this, we present the main algorithm which involves two phases:

Top-down phase: First, we use a **top-down** strategy, iterating through the top half of the layers. For each layer $l \in (1 \dots \lfloor L/2 \rfloor)$ (starting from the root and moving downward), we cluster the documents using **clusterer**.

For the first layer $L = 1$, we start by dividing the entire collection of papers into k_1 clusters at the top level using their embeddings. For the second level, we process each top-level cluster independently, i.e., within each top-level cluster, we take only the papers belonging to that cluster and apply **clusterer** to their embeddings to divide them into subgroups.

This process continues recursively: for subsequent layers, clustering operates on the data points within each cluster. We run as many clustering algorithms as there are clusters in the previous layer. The allocation of the total cluster count across different top-level clusters is determined proportionally based on the number of papers in each top-level cluster. If a top-level cluster contains a larger proportion of papers, it will be allocated more subclusters

from the total cluster count. This proportional allocation ensures that areas with higher paper density receive finer-grained clustering.

This recursively subdivides each cluster at level l into multiple clusters at level $l + 1$. This top-down phase continues until we reach level $\lfloor L/2 \rfloor$. Alongside clustering, we also use the **summarizer** to generate abstracted summaries for each of the clusters. The generated cluster description follows the same structure or style as the input descriptions. For example, if the inputs are statements about problem categories, the output from **summarizer** is also in the same style, but more abstract.

Bottom-up phase: In the second phase, we switch to a **bottom-up** strategy to construct the remaining levels ($\lfloor L/2 \rfloor + 1$ through L). To form clusters for bottom-level (layer L), we apply **clusterer** to the paper embeddings within each sub-cluster within level- $\lfloor L/2 \rfloor$ (the lowest level clustering obtained from top-down approach). We then use the **summarizer** to create an abstracted description for each cluster.

We repeat this process for all layers from L to $\lfloor L/2 \rfloor + 1$. To build layer l , we start by embedding the generated cluster summaries from the level below $l - 1$ using **embedder**, similar to how we embedded the papers. We then run the clustering **clusterer** on these new embeddings and generate abstracted summaries for the clusters to group these summaries into higher-level clusters. This bottom-up aggregation continues until we connect with the previously constructed level $\lfloor L/2 \rfloor$ clusters.

Rationale behind the hybrid design: The hybrid approach merges the strengths of top-down and bottom-up strategies. A bottom-up method may create less coherent top-level clusters. The top-down approach ensures high-quality top-level clusters but doesn’t utilize the abstracted summaries from **summarizer** used by bottom-up clustering. By combining both methods, the hybrid design achieves robust and effective clustering. Our empirical results in §5.4 demonstrate this approach’s strength by balancing quality and scalability.

4.2 Top-down and Bottom-up Baselines

We explore special cases of SCYCHIC: one that employs only the top-down approach and another that uses only the bottom-up approach. These variations allow us to assess the individual strengths and weaknesses of each strategy. We assess these specific cases in §5.4.

4.3 Pure LLM-based Baselines

We introduce baselines that heavily utilize LLM calls, based on the hypothesis that LLMs can make high-quality local decisions, collectively forming a robust global structure. The potential cost here is the need to make *many* LLM calls. We refer to these baselines as FLMSCI (pronounced “flimsy”) and present two variants below.

Initializing a Seed Hierarchy: The first step involves creating a seed hierarchy, starting with the hierarchy of sciences from the Wikipedia page on branches of science.² We made several adjustments to this hierarchy, detailed in §D, where the resulting seed hierarchy is also included.

Algorithm 1 SCYCHIC algorithm

Require: Set of papers $P = \{p_1, p_2, \dots, p_n\}$, **embedder**, **clusterer**, **summarizer**, num of layers L , target cluster sizes $(k_1, k_2, \dots, k_L)$

- 1: **Initialization:** For each paper $p_i \in P$, embed their selected components to form $\mathbb{R}^{d \times |C|}$.
- 2: **for** layer $l = 1$ to $\lfloor L/2 \rfloor$ **do** ▷ Top-down phase
- 3: **if** $l = 1$ **then**
- 4: Apply **clusterer** to divide papers into k_1 clusters
- 5: **else**
- 6: **for** each cluster from layer $l - 1$ **do**
- 7: Apply **clusterer** to divide into subclusters
- 8: Use **summarizer** to generate summaries for each cluster
- 9: **for** each cluster κ at level $\lfloor L/2 \rfloor$ **do** ▷ Bottom-up phase
- 10: **for** layer $l = L$ to $\lfloor L/2 \rfloor + 1$ **do**
- 11: **if** $l = L$ **then**
- 12: Collect the embeddings of papers within κ .
- 13: **else**
- 14: Apply **embedder** on summaries of cluster $l + 1$
- 15: Apply **clusterer** to form higher-level clusters
- 16: Use **summarizer** to generate abstracted summaries
- 17: **return** Hierarchical structure

²https://en.wikipedia.org/wiki/Branches_of_science

FLMSCI (parallel): Adding Many Items in Parallel with Few Prompts: This approach involves adding items to the seed hierarchy in parallel. First, gather all unique items extracted from different papers and create batches of 100 items each (batching because all of these items would not fit within the context window of LLM). Then, implement a multi-threaded program where each thread adds a batch of keywords to a clone of the seed hierarchy. Finally, merge (with a Python program, not an LLM call) all these cloned trees into a single comprehensive structure.

FLMSCI (incremental): Expand the tree by incrementally adding items one at a time: This approach involves an iterative, layer-by-layer prompt that navigates the tree starting from the root node and performs specific actions. By default, the supported actions include: (a) *Go down*: select one of the nodes to traverse down to a lower layer in the tree; (b) *Add sibling*: Insert a new node at the same level; (c) *Make parent*: Create a new parent node; (d) *Discard*: Ignore the item, if no suitable location is found. This prompt are shown in Fig.11.

The available actions vary depending on the tree’s current position. To prevent placing detailed items too high in the tree (above layer 3), we hide the actions that involve nodes (b) and (c). This restriction provides more control and reduces errors, as observed in pilot experiments. If the traversal reaches a child node, we hide the option to move further down (a). Another issue identified was frequent mistakes in earlier layers, likely due to broad labels where multiple categories could fit. To address this, we provide more context for the first layer by replacing label strings with descriptive definitions, as shown in Fig.10.

4.4 The Computational Complexity of Each Approach

An important bottleneck for building a hierarchy is the number of LLM calls. Let C denote the total number of contributions (§3.2), b the branching factor, and L the maximum depth set to $L = O(\log_b C)$ for a near-balanced tree (§3.3). For our proposed algorithm, SCY-

Approach	# of LLM calls
SCYCHIC	$O(C/b)$
FLMSCI (parallel)	$O(C/l)$
FLMSCI (incremental)	$O(C \log_b C)$

CHIC, both the bottom-up and top-down variants incur a total of $O\left(\frac{C}{b} + \frac{C}{b^2} + \dots + \frac{C}{b^L}\right) = O(C/b)$ LLM calls. This is because at each layer of the hierarchy, the number of nodes decreases geometrically by a factor of b . For the LLM-based baselines presented in §4.3: FLMSCI (parallel) involves processing contributions in batches of size l (in experiments we use $l = 100$). Consequently, the total number of LLM calls is given by $O(C/l)$. In FLMSCI (incremental), each of the C items is inserted into the hierarchy by traversing from the root to a leaf. In a tree of depth $L = O(\log_b C)$, this traversal requires $O(\log_b C)$ steps per item, giving us a total number of LLM calls: $O(C \log_b C)$. In the experiments, we report the number of LLM calls for each approach where we see that the gap between $O(C \log_b C)$ and $O(C/b)$ is quite drastic (61K vs 322, in Table 1).

5 Experimental Setup and Results

We provide further context on our experimental setup: the collection of papers with diverse topics which we aim to place on a hierarchy (§5.1), and our evaluation setup (§5.2).

5.1 Collection of Science Papers

We compile a set of science papers to used in our study. Our primary analysis is conducted on a relatively small collection of roughly 2K papers, spanning domains such as computer science, neuroscience, biology, oceanography, and their interdisciplinary connections. We refer to this collection as **SciPile**. This smaller set allows for rapid iteration over the data, enabling us to test various algorithms and assess their scalability. These papers are used to validate our approach in generating high-quality hierarchies. To evaluate whether our methods can scale to larger collections and create a more comprehensive hierarchy, we curated an expanded dataset of 10K papers, referred to as **SciPileLarge**. Details on our collection and filtering is in §F.

5.2 Evaluation as Utilization

In an ideal world, we would compare the inferred hierarchy to a gold standard, but such data doesn’t exist, and scientific literature constantly evolves. Therefore, we need an evaluation framework independent of a fixed ground truth.

We propose assessing the quality and utility of generated hierarchies from a *utilization* perspective, focusing on their ability to aid navigation and identify relevant content. We use an LLM-based agent to locate target information within the hierarchy, performing tree traversal and measuring accuracy. Evaluation occurs at each level and across the entire hierarchy to determine overall success. A superior hierarchy enhances information discovery, capturing conceptual relationships among subtopics. While we evaluate based on its ability to locate information, this doesn’t restrict the hierarchy’s other uses.

Key considerations are (a) selecting suitable queries and (b) choosing the most accurate LM models. For (a), we sample a paper (title and abstract). We also considered using questions derived from the paper’s content to simulate typical information-seeking behavior. However, in our pilot studies, we found that results from both methods generally correlate, so we opted for the former. For (b), we use Qwen2.5-32b-instruct due to its strong instruction-following capabilities. In pilot experiments, Qwen demonstrated the highest consistency compared to other open-weight models and GPT-4 (see §B).

Starting at the top of the hierarchy, we present the evaluator (LLM) with our query and the clusters’ contributions (see prompt in Fig.2). The LLM must select the most appropriate cluster. We proceed to the next hierarchy level only if the model correctly identifies the cluster containing the target paper. At each subsequent level, we use the same query, presenting only the subclusters within the previously selected correct cluster. This process continues until the model successfully navigates to the level containing the target paper.

We report two metrics: Strict Accuracy (**Strict-Acc**), which measures how often the LLM correctly identifies the exact node, and Layer-1 Accuracy (**L1-Acc**), which measures how often the LLM correctly identifies the subtree containing the query.

Method	Accuracy (%)		LLM Cost		Hierarchy Structure				
	Strict-Acc $\uparrow$	L1-Acc $\uparrow$	Avg. # of Input Tokens $\downarrow$	# of Calls $\downarrow$	Max Depth	Avg Depth	Avg Bran. Factor	Max Bran. Factor	# of Items
<i>Contributions type: Topic</i>									
SCYCHIC	14.9 $\pm$ 2.7	65.7 $\pm$ 4.4	5017	322	3	3	40.9	128	11k
↳ Top-down	14.9 $\pm$ 2.7	62.5 $\pm$ 7.4	6440	322	3	3	40.9	104	11k
↳ Bottom-up	13.9 $\pm$ 5.3	54.4 $\pm$ 12.7	3988	322	3	3	40.9	119	11k
↳ FLMSCI (par)	4.0 $\pm$ 2.8	32.0 $\pm$ 6.3	8896	226	9	6.2	13.9	734	9.9K
↳ FLMSCI (inc)	18.0 $\pm$ 5.3	91.0 $\pm$ 4.0	4040	61K	14	7.7	3.6	704	10.4K

Table 1: Evaluation results of SCYCHIC, FLMSCI (**parallel**) and FLMSCI(**incremental**) when using *Topic* as contribution type. “Bran.” stands for “Branching”. All methods show poor Strict-Acc ($\leq 18.0\%$), highlighting the challenging nature of the task. On one hand, FLMSCI (inc) achieves the highest accuracy, showing the feasibility of building hierarchies by pure LLM-based methods. However, it incurs substantial computational costs, about $200\times$ higher than other methods. In contrast, SCYCHIC offers a balanced performance profile with competitive accuracy (14.9% Strict-Acc, 65.7% L1-Acc) while maintaining significantly lower computational requirements.

5.3 Experiment Design

Following the proposed evaluation protocol, we conduct three sets of experiments to compare our method (SCYCHIC, §4.1) with baselines (§4.2, 4.3). The hyperparameters of SCYCHIC are in §G. The experiments are organized as follows: (1) We first compare all approaches on “topic” contributions, the simplest form of contributions. (2) Next, we assess the methods on more complex contributions (problem definition, results, and solutions).

Method	Accuracy (%)		LLM Cost		Hierarchy Structure		
	Strict-Acc $\uparrow$	L1-Acc $\uparrow$	Avg. # of Input Tokens $\downarrow$	# of Calls $\downarrow$	Depth	Avg Branching Factor	Max Branching Factor
<i>Contributions type: Problem Statement</i>							
SCYCHIC	51.1 $\pm$ 3.8	81.7 $\pm$ 2.6	2624				20
↳ Top-down	49.0 $\pm$ 3.7	80.3 $\pm$ 2.7	2953	322	3	7.1	18
↳ Bottom-up	45.9 $\pm$ 5.0	69.3 $\pm$ 8.1	2177				16
<i>Contributions type: Solution Statement</i>							
SCYCHIC	48.8 $\pm$ 6.1	82.3 $\pm$ 1.1	2343				16
↳ Top-down	45.9 $\pm$ 5.5	79.2 $\pm$ 3.4	2521	322	3	7.1	19
↳ Bottom-up	14.8 $\pm$ 4.4	22.0 $\pm$ 5.7	1990				14
<i>Contributions type: Results Statement</i>							
SCYCHIC	46.4 $\pm$ 5.2	76.4 $\pm$ 6.9	2654				16
↳ Top-down	45.2 $\pm$ 4.5	71.4 $\pm$ 3.7	2718	322	3	7.1	16
↳ Bottom-up	40.0 $\pm$ 10.7	64.0 $\pm$ 8.9	2210				13

Table 2: Evaluation results of SCYCHIC and the corresponding baselines. SCYCHIC maintains the highest accuracy and a relatively small variance across all contribution types, which proves the rationale behind our hybrid design. Among three contribution types, *problem statement* contribution type hierarchies have the highest **Strict-Acc** (51.1%) and near first **L1-Acc**(81.7%), indicating this contribution type contributes most for hierarchy construction.

(3) Finally, the third set of experiments evaluates scalability on **SciPileLarge**. Due to the computational cost (number of API calls), we evaluate the LLM-based baselines only for the first set.

5.4 Empirical Findings

Tables 1, 2, and 3 present our experimental results. They also include *LLM Cost*, detailing the average number of input tokens and LLM calls, and *Hierarchy Structure*, which provides statistics on depth and branching factors (i.e., the number of children per node).

SCYCHIC outperforms its special-cases baselines. From Table 1, we observe that SCYCHIC significantly outperforms the top-down and bottom-up baselines in Level-1 accuracy while achieving comparable Strict-Acc. Similar outcomes are also seen in Table 2 for contribution types. For instance, when on “solution” contributions, SCYCHIC outperforms the top-down baseline by 2.9% in Strict-Acc and 3.1% in L1-Acc, demonstrating the effectiveness of our method. Moreover, these strong performances are achieved with a comparable number of tokens and LLM calls as the baselines, making SCYCHIC computationally efficient.

LLM-based baselines are much costlier than SCYCHIC. Although the FLMSCI baseline outperforms SCYCHIC in both accuracies by a small margin, it requires an *prohibitively* large number of calls, rendering it too costly for large-scale scientific literature. Despite its good quality, FLMSCI (incremental) will not scale.

SCYCHIC scales to larger paper corpus. For our scalability experiments, we evaluated SCYCHIC on our larger 10K paper dataset **SciPileLarge**, using the *problem statement* contribution type. Due to the significant increase ($\times 5$) in corpus size, we implemented a 4-layer hierarchy instead of the 3-layer structure used previously. Notably, SCYCHIC achieved even higher L1-Acc (86.5%) on **SciPileLarge** compared to our smaller dataset **SciPile**. This improvement likely stems from the enhanced quality of our expanded dataset, which has more strict filtering mechanisms. While the Strict-Acc showed a minor decrease compared to results on **SciPile**, it remained at a satisfactory level. Collectively, these results provide compelling evidence that our method scales successfully to substantially larger paper corpora.

Method	Accuracy (%)		LLM Cost		Hierarchy Structure			
	Strict-Acc $\uparrow$	L1-Acc $\uparrow$	Avg. # of Input Tokens $\downarrow$	# of Calls $\downarrow$	Depth (Max/Avg)	Avg Bran. Factor	Max Bran. Factor	# of Items
<i>Contributions type: Problem Statement</i>								
SCYCHIC	43.7 $\pm$ 6.5	85.8 $\pm$ 4.2	7451	1572	4	8	26	10K
$\downarrow$ Top-down	41.5 $\pm$ 8.2	86.5 $\pm$ 5.6	8990	1572	4	8	30	10K
$\downarrow$ Bottom-up	26.2 $\pm$ 5.4	41.9 $\pm$ 4.0	5924	1572	4	8	26	10K

Table 3: Evaluation results of SCYCHIC when building hierarchies on **SciPileLarge** (dataset with 10K papers). Compared to **SciPile** (the 2K dataset), our method shows a slight decrease in Strict-Acc, while even achieves higher L1-Acc. This fully demonstrates the feasibility of our method on the scaling dataset.

6 Discussion and Conclusion

Future applications: Our work paves the way for several promising research directions. One potential application is to leverage the constructed hierarchies as tools for exploratory analysis in various scientific fields. These hierarchies can help academic institutions and funding agencies identify emerging trends and under-explored areas, and they can be refined for domain-specific analyses that capture the nuances of individual disciplines. This approach not only enriches our understanding of scientific progress but also offers a new perspective on organizing the vast landscape of scholarly work.

Limitations: Although we evaluated our pipeline on 10K papers, this is still far from the true scale of scientific literature. We hope future work will enhance our approach to handle more realistic scales. Additionally, while our evaluation framework shows potential for efficient information discovery, it may have its own weaknesses and biases. Integrating human verification into the assessment process could help ensure the quality and reliability of the inferred hierarchies.

Conclusions: In summary, we have motivated SCIENCE HIERARCHOGRAPHY, large-scale hierarchical summarization of scientific literature, offering a fresh perspective on how research efforts are organized and distributed. Our method, SCYCHIC, effectively combines LLMs with faster procedures to balance quality and scalability. Looking ahead, we hope our work offers researchers a more intuitive way to navigate the complex landscape of scholarly and enhances the allocation of resources in academia.

Ethics Statement

In our work, all data and models are accessed via licenses that grant us free and open access for research purposes. Expert annotations are provided by the paper’s authors, who have contributed their efforts without compensation. We have not observed any harmful content in either the scholarly papers or the content generated by LLMs. On the other hand, since our resulting hierarchy reflects the distribution of scientific efforts across various fields, it offers a detailed map of where research activity is concentrated and where it is lacking. This nuanced view can guide decision-makers—such as government agencies and academic institutions—in making more informed choices about resource allocation. By highlighting underexplored yet promising areas alongside well-established fields, the hierarchy helps ensure that funding, support, and strategic initiatives are distributed more equitably. Ultimately, this balanced approach can foster innovation and drive progress in areas that might otherwise be overlooked, leading to a more inclusive and socially beneficial advancement of science.

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A Additional Related Work

We include additional related work here, because of the space limitation in the main text.

Structured knowledge in LLMs: Prior work has explored how LLMs internalize hierarchical knowledge. For example, [He et al. \(2024\)](#); [Lovón-Melgarejo et al. \(2023\)](#); [Park et al. \(2025\)](#) extend the linear representation hypothesis to reveal that LLMs encode categorical concepts as polytopes, with hierarchical relationships reflected as orthogonal directions. Other works such as [Wolfman et al. \(2024\)](#) and [Budagam et al. \(2024\)](#) examine the benefits of explicit hierarchical syntactic structures and prompting frameworks for guiding LLM performance, while [Moskvoretskii et al. \(2024\)](#) and [Hu et al. \(2024a\)](#) focus on constructing and materializing large-scale structured knowledge bases about entities and events. In line with the same aspirations, our work explores the use of hierarchical structures to organize scientific literature.

Structured knowledge representation: Understanding and organizing knowledge is a fundamental pursuit in both artificial and human intelligence ([Dahlberg, 1993](#)). Abstraction hierarchies, such as WordNet for lexical semantics ([Miller, 1995](#)), ConceptNet for common-sense reasoning ([Speer et al., 2017](#)), and Probase for large-scale concept representation ([Wu et al., 2012](#)), have proven to be powerful tools for structuring information. Similarly, modern tabular reasoning leverages structured representations to facilitate systematic inference and knowledge retrieval, demonstrating that such structure remains crucial ([Wang et al., 2024](#)).

B Evaluation Framework

We provide more context on our evaluation. As discussed in §5.2, we use randomly-sampled papers (title/abstract) as a query. The evaluator LLM goes through the hierarchy, starting from the root node and iteratively selects the relevant nodes to traverse. The prompt for each decision is shown in Fig.2.

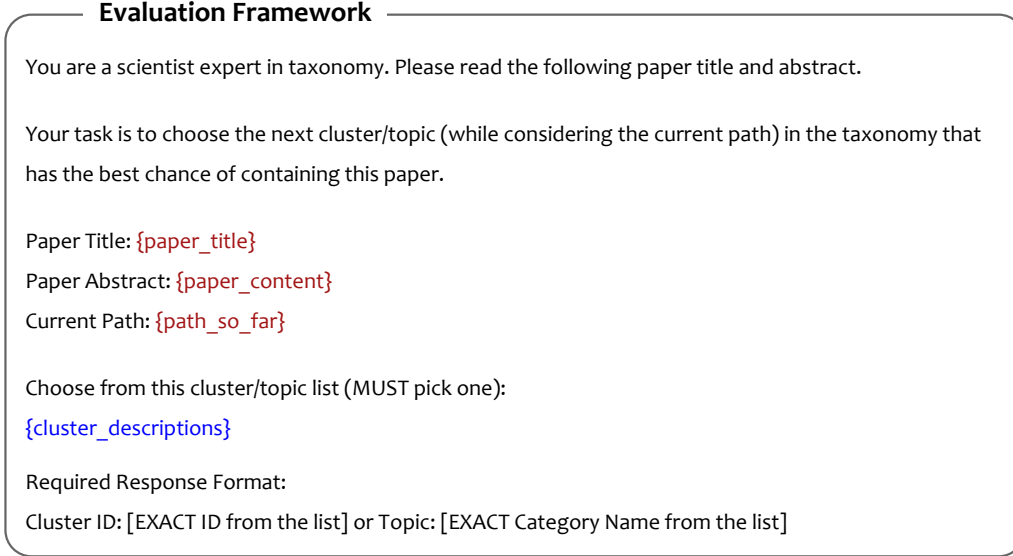


Figure 2: Prompt used for Evaluation

One question is, **which LLM should we use for evaluation?** We chose Qwen2.5-32b-instruct for its strong instruction-following capabilities. In pilot experiments, Qwen showed a high consistency against GPT4 score, compared to other open-weight models. Here’s a summary of that experiment: We evaluated one of the hierarchies produced by SCYCHIC using different models, including GPT-4. Assuming GPT-4 has the highest accuracy, we sought alternative models with the greatest consistency against it, as frequent evaluations with GPT4 are costly. Fig.4 presents the results. As it can be observed, Llama has the highest agreement, but we suspect bias since the hierarchy was also constructed with Llama. To avoid this, we selected the next best model, Qwen2.5-32b-instruct, for evaluation.

Evaluator LLM	Agreement with GPT4
GPT-3.5	39.6
GPT4-mini	59.2
Gemma3-24b-it	62.1
Qwen2.5-32b-instruct	66.5
Llama 3.3 70B	72.4

Table 4: Agreement of different evaluator LLMs against GPT4.

C Extracting Paper Contributions

Below are prompts and examples for extracting different contributions (*problem*, *solution*, *result* and *topics*) from papers' titles and abstracts. we utilize the GPT-4o model (gpt-4o-2024-08-06) to generate all contribution extractions along with detailed rationales explaining the extraction decisions.

C.1 Prompt for Extracting *Problem/Solution/Result* Contributions

We use the prompt below to extract contributions from the paper's title and abstract. After finishing the extraction, the three contributions will be saved into the original json file.

Contributions Extraction from Paper

Consider the following following paper:

Title: {title}

Abstract: {abstract}

Extract the relevant content of the above abstract into the following JSON structure. For certain fields that the information is not found in the abstract, leave them empty (empty string).

```
{
  "problem": {
    "overarching problem domain": "",
    "challenges/difficulties": "",
    "research question/goal": "",
    "novelty of the problem": "",
    "knowns or prior work": "",
  },
  "solution": {
    "overarching solution domain": "",
    "solution approach": "",
    "novelty of the solution": "",
    "knowns or prior work": "",
  },
  "results": {
    "findings/results": "",
    "potential impact of the results": "",
  },
}
```

Figure 3: Prompt for extracting *Problem/Solution/Result* contributions

C.2 Prompt for Extracting *Topic* Contributions and Rationales

This section has the prompt of generating topics and rationales from papers given their titles and abstracts. The prompt provides the model with a system role instruction that describes the task, title, and abstract, and also an example to get the specified output format.

Topics and Rationales Generation

You are an experienced scientist who is going to read and review research papers.

Paper Title: {title}

Paper Abstract: {abstract}

Read the above given Title and Abstract for a research paper and
Generate topics that are represented in the given Title and Abstract.
Example output format:

```
```json
{
 "topics": [
 {
 "topic": "Entity Taxonomy Creation",
 "rationale": "The research focuses on generating a comprehensive entity taxonomy using LLMs."
 },
 {
 "topic": "Iterative Prompting Techniques",
 "rationale": "Highlights the use of iterative prompting to refine entity classifications."
 },
 {
 "topic": "GPT-4 and GPT-4 Turbo",
 "rationale": "Explores the capabilities of these advanced LLMs in taxonomy development."
 },
 {
 "topic": "Information Extraction",
 "rationale": "Demonstrates applications like relation and event argument extraction."
 },
 {
 "topic": "Computational Linguistics",
 "rationale": "Emphasizes contributions to AI-related and linguistic computational tasks."
 }
]
}
```

Figure 4: Prompt of *Topic* and *Rationale* Generation

## C.3 Examples for *Problem/Solution/Result/Topic* contributions extracted from papers

Below we show examples of paper titles and abstracts, and different contributions (*Problem/-Solution/Result/Topic*) we extract by language model.



**Problem/Solution/Result/Topic contributions from scientific papers****Title:** Sixfold excitations in electrides

**Abstract:** Due to the lack of full rotational symmetry in condensed matter physics, solids exhibit new excitations beyond Dirac and Weyl fermions, of which the sixfold excitations have attracted considerable interest owing to the presence of maximum degeneracy in bosonic systems. Here, we propose that a single linear dispersive sixfold excitation can be found in the electride  $\text{Li}_{12}\text{Mg}_3\text{Si}_4$  and its derivatives. The sixfold excitation is formed by the floating bands of elementary band representation  $A@12a$  originating from the excess electrons centered at the vacancies (i.e., the 12a Wyckoff sites). There exists a unique topological bulk-surface-edge correspondence for the spinless sixfold excitation, resulting in trivial surface "Fermi arcs" but topological hinge arcs. All gapped  $k_z$  slices belong to a two-dimensional higher-order topological insulating phase, which is protected by a combined symmetry  $T S_{4z}$  and characterized by a quantized fractional corner charge  $Q_{\text{corner}} = 3|e|/4$ . Consequently, the hinge arcs are obtained in the hinge spectra of the  $S_{4z}$ -symmetric rod structure. The state with a single sixfold excitation, stabilized by both nonsymmorphic crystalline symmetries and time-reversal symmetry, is located at the phase boundary and can be driven into various topologically distinct phases by explicit breaking of symmetries, making these electrides promising platforms for the systematic studies of different topological phases.

Contribution - Problem Statement	Contribution - Solution Statement	Contribution - Result Statement
<pre>{   "overarching_problem_domain":     "Condensed matter physics",   "challenges/difficulties":     "Lack of full rotational symmetry     in solids leading to new excitations     beyond Dirac and Weyl fermions",   "research_question/goal":     "Investigate sixfold excitations     in electrides" }</pre>	<pre>{   "overarching_solution_domain":     "Electrides and topological phases",   "solution_approach":     "Propose that a single linear     dispersive sixfold excitation can     be found in the electride     <math>\text{Li}_{12}\text{Mg}_3\text{Si}_4</math> and its derivatives",   "novelty_of_the_solution":     "Unique topological bulk-surface-edge     sixfold excitation" }</pre>	<pre>{   "findings/results":     "The sixfold excitation is formed by     floating bands of elementary band     representation <math>A@12a</math>. All gapped     <math>k_z</math> slices belong to two-dimensional     higher-order topological insulating     phase, characterized by a quantized     fractional corner charge <math>Q_{\text{corner}} = 3 e /4</math>.     Hinge arcs are obtained in the hinge     spectra of the <math>S_{4z}</math>-symmetric rod     structure.",   "potential_impact_of_the_results":     "Electrides are promising platforms     for systematic studies of different     topological phases." }</pre>

**Contribution - Topic:** 'Electrides', 'Electrides in Condensed Matter Physics', 'Higher-Order Topological Insulators', 'Nonsymmorphic Symmetries', 'Sixfold Excitation in Solids', 'Sixfold Excitations', 'Symmetry Breaking in Topological Materials', 'Topological Bulk-Surface-Edge Correspondence', 'Topological Phase Transitions', 'Topological Phases in Condensed Matter Physics', 'Topological Properties'

**Title:** The Tin Pest Problem as a Test of Density Functionals Using High-Throughput Calculations

**Abstract:** At ambient pressure tin transforms from its ground-state semi-metal  $\alpha$ -Sn (diamond structure) phase to the compact metallic  $\beta$ -Sn phase at  $13 \bullet \text{C}$  (286K). There may be a further transition to the simple hexagonal  $\gamma$ -Sn above 450K. These relatively low transition temperatures are due to the small energy differences between the structures,  $\approx 20$  meV/atom between  $\alpha$ - and  $\beta$ -Sn. This makes tin an exceptionally sensitive test of the accuracy of density functionals and computational methods. Here we use the high-throughput Automatic-FLOW (AFLOW) method to study the energetics of tin in multiple structures using a variety of density functionals. We look at the successes and deficiencies of each functional. As no functional is completely satisfactory, we look Hubbard U corrections and show that the Coulomb interaction can be chosen to predict the correct phase transition temperature. We also discuss the necessity of testing high-throughput calculations for convergence for systems with small energy differences.

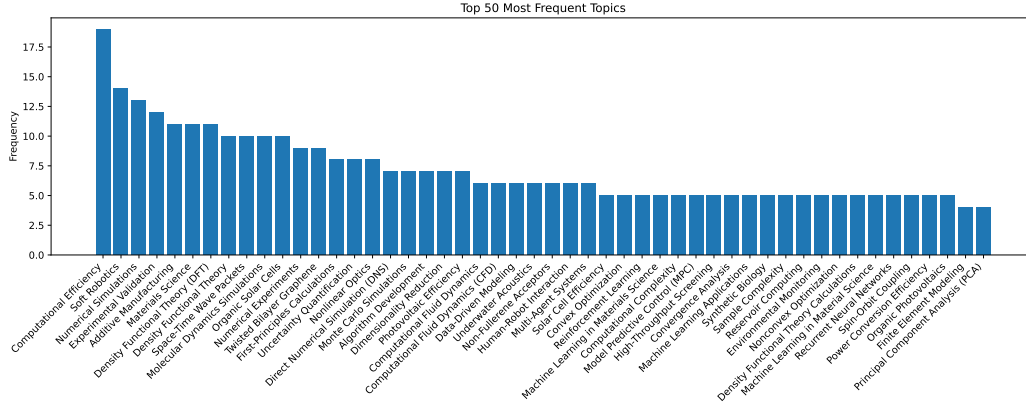
Contribution - Problem Statement	Contribution - Solution Statement	Contribution - Result Statement
<pre>{   "overarching_problem_domain":     "Density functionals and computational     methods for phase transitions in     materials.",   "challenges/difficulties":     "Small energy differences between     phases of tin make it a sensitive     test for the accuracy of density     functionals.",   "research_question/goal":     "To study the energetics of tin in     multiple structures using a variety     of density functionals and assess     their accuracy." }</pre>	<pre>{   "overarching_solution_domain":     "High-throughput computational     methods and density functional     theory.",   "solution_approach":     "Using the high-throughput     Automatic-FLOW (AFLOW) method     to study tin's energetics with     various density functionals.",   "novelty_of_the_solution":     "Application of Hubbard U     corrections to improve predictions     of phase transition temperatures." }</pre>	<pre>{   "findings/results":     "No functional is all satisfactory,     but Hubbard U corrections can be chosen     to predict the correct phase     transition temperature.",   "potential_impact_of_the_results":     "Improved accuracy in predicting     phase transitions in materials     with small energy differences." }</pre>

**Contribution - Topic:** 'Convergence Testing in Computational Simulations', 'Density Functional Theory (DFT) Accuracy', 'High-Throughput Computational Methods', 'Hubbard U Corrections', 'Tin Phase Transitions'

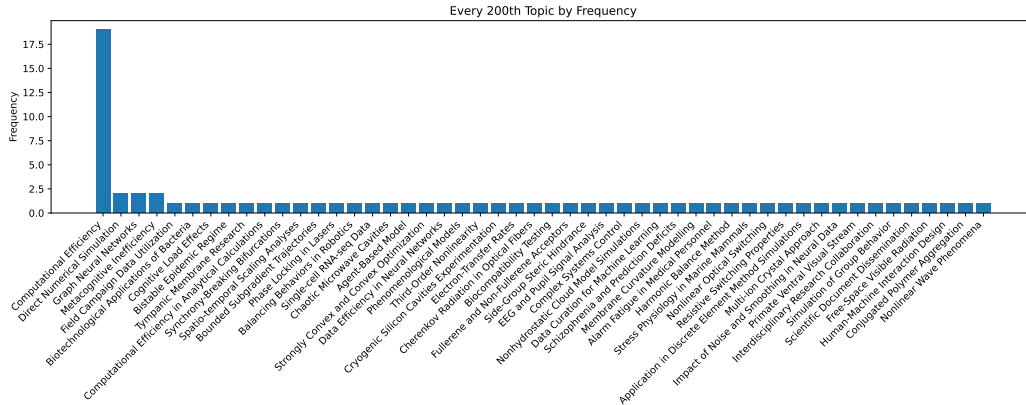
Table 5: Examples of extracted *problem/solution/result/topic* contributions from scientific paper abstracts.

## C.4 Distribution of Extracted Topics

This section shows the distribution of various topics extracted from the papers based on frequency. This gives us an idea of what kind of topics were extracted.



(a) Top-50 topics by frequency in decreasing order



(b) Sampled topics (every 200)

Figure 5: Distribution of topics extracted from SciPile: (a) Top-50 topics, (b) Every 200 topics. Refer 3.2 for more information

## D Compiling a Seed Hierarchy

As we discuss in §4.3, we make a few adjustments to the seed hierarchy that we obtain from Wikipedia. Specifically:

1. We added “Theoretical Computer Science” and “Information Theory” as separate branches under “Formal Sciences” due to their unique characteristics;
2. We moved “Astronomy” under “Physical Science”;
3. “Astronomy”, “Geology” and “Oceanography” are listed under “Earth Science” but since these topics are not specific to early, we move them up one layer so that they’re directly under “Physical Science”; The Wikipedia article groups Geology, Oceanography, and Meteorology under ;
4. We added “History” as a branch under “Social Sciences”;
5. We included “Cell Biology” and “Genetics” under “Biological Sciences” as they were relevant and their inclusion would only help in better hierarchy creation.

These modifications aim to refine the hierarchy, ensuring it accurately reflects the distinct areas within each scientific domain. The resulting hierarchy is included in Fig.6.

```

1 {
2 "Science":{
3 "Formal Sciences":{
4 "Logic":{},
5 "Mathematics":{},
6 "Statistics":{},
7 "Computer Science":{},
8 "Information Theory":{},
9 "Systems Theory":{},
10 "Decision Theory":{}
11 },
12 "Natural Sciences":{
13 "Physical Science":{
14 "Physics":{
15 "Classical Mechanics":{},
16 "Thermodynamics and statistical mechanics":{},
17 "Electromagnetism and photonics":{},
18 "Relativity":{},
19 "Quantum Mechanics":{},
20 "Atomic and molecular physics":{},
21 "Condensed matter physics":{},
22 "Optics and acoustics":{},
23 "High energy particle physics":{},
24 "Nuclear physics":{},
25 "Cosmology":{},
26 "Interdisciplinary Physics":{}
27 },
28 "Chemistry":{
29 "Physical Chemistry":{},
30 "Organic Chemistry":{},
31 "Inorganic Chemistry":{},
32 "Analytical Chemistry":{},
33 "Biological Chemistry":{},
34 "Theoretical Chemistry":{},
35 "Interdisciplinary Chemistry":{}
36 },
37 "Earth Science":{},
38 "Astronomy":{},
39 "Geology":{},
40 "Oceanography":{},
41 "Meteorology":{}
42 },
43 "Biological Sciences":{
44 "Biochemistry":{},
45 "Cell Biology":{},
46 "Genetics":{},
47 "Ecology":{},
48 "Microbiology":{},
49 "Botany":{},
50 "Zoology":{}
51 }
52 },
53 "Social Sciences":{
54 "Anthropology":{},
55 "Economics":{},
56 "Political Science":{},
57 "Sociology":{},
58 "Psychology":{},
59 "Geography":{},
60 "History":{}
61 }
62 }
63 }

```

Figure 6: The seed hierarchy used by our FLMSCI baselines. See §D for details.

## E FLMSCI: LLM-based Baselines

This section includes the pipeline and prompts used for FLMSCI (parallel) and FLMSCI(incremental).

### E.1 Pipeline for FLMSCI (parallel)

This section demonstrates the pipeline used for FLMSCI (par) right from extracting topics and rationales to obtaining a final taxonomy with papers. (Refer to §4.3 for more information).

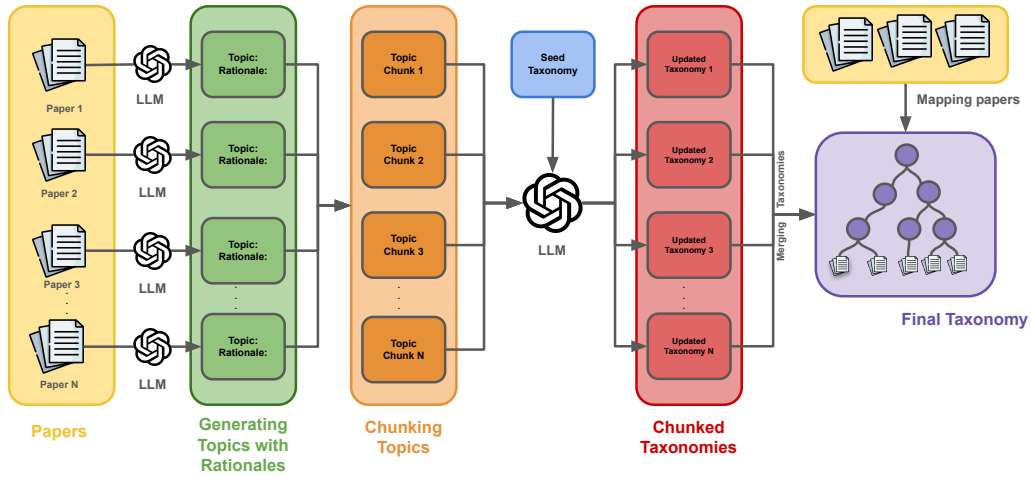


Figure 7: Pipeline for of FLMSCI (parallel).

## E.2 Prompt for FLMSCI (parallel)

This prompt guides a large language model (LLM) to expand an existing scientific taxonomy - the seed taxonomy (Refer [D](#)) by adding a set of new topics in a logical and consistent manner. With a clear list of instructions it ensures accurate placement and also preserves the original structure. This prompt was used with Llama-3.3-70B-Instruct. (Refer [§4.3](#) for more information.)

### FLMSCI (parallel) Prompt

You are a scientific domain expert. You have an initial "seed taxonomy" of scientific concepts and a list of new topics to integrate into that taxonomy. Please carefully analyze these new topics and update the seed taxonomy so that:

1. You must retain the current structure of the seed taxonomy and respect all existing categories.
2. Place each and every topic from the list given below.
2. You are free to add new branches or subcategories only where necessary to fit the new topics in a consistent, hierarchical ("is-a") manner.
3. Each topic from the list must appear exactly once. Do not introduce any new topics beyond those in the list.
4. Ensure each new topic is placed under the correct parent concept based on its semantic meaning or specialization level.
5. Return your updated taxonomy as valid JSON, containing both the original seed hierarchy and the newly incorporated topics.

Below is your seed taxonomy (in JSON). Make sure to preserve its structure as much as possible:

`{seed_taxonomy}`

Here is the list of new topics that must be integrated:

`{topics_chunk}`

Focus on logical placement of each term to maintain an accurate scientific hierarchy.

Figure 8: Prompt of FLMSCI (incremental) pipeline



### E.3 Demonstration of actions for FLMSCI (incremental)

This section demonstrates the various actions (add sibling, make parent, go down and discard) that are available for the LLM to take at various levels of taxonomy building. Refer §4.3 for more information

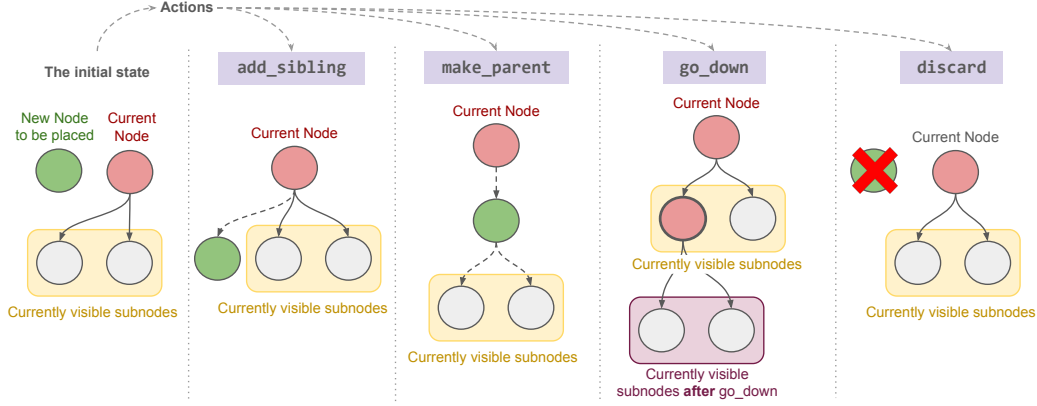


Figure 9: Actions for FLMSCI (incremental)

### E.4 Prompt for FLMSCI (incremental)

This prompt is used to place new scientific topics into an existing seed taxonomy (Refer §D) incrementally. The model evaluates multiple possible actions based on the available action options. The prompt clearly instructs its priorities explicitly to give a hint to the model. The example usage and example output format help to get the response in a valid format. This prompt was used for Llama-3.3-70B-Instruct.

```

1 SUBNODE_DESCRIPTIONS = {
2 "Formal Sciences": "Focuses on abstract systems and formal methodologies grounded in
3 logic, mathematics, and symbolic reasoning. Provides theoretical frameworks (e.g.,
4 statistics, computer science, systems theory) used to model and solve problems
5 across empirical disciplines and technology.",
6 "Natural Sciences": "Investigates the physical universe and living organisms through
7 empirical observation, experimentation, and theoretical analysis. Includes physical
8 sciences (e.g., physics, chemistry, astronomy) and biological sciences (e.g.,
9 genetics, ecology) to uncover fundamental laws and processes governing nature.",
10 "Social Sciences": "Studies human behavior, societies, and institutions using
11 qualitative and quantitative methods. Encompasses disciplines like psychology,
12 economics, and political science to analyze cultural, economic, and social
13 interactions within historical and geographic contexts."
14 }

```

Figure 10: Descriptive statement used for contextualizing layer-1 items in the seed hierarchy, used in FLMSCI (incremental). See §4.3 for broader context.

```

1 You are building a scientific topics based hierarchy.
2
3 Path traced until now: {current_path}
4 Subnode options available at this level:
5 subnodes = [{subnodes}]
6 New topic: "{new_topic}"
7
8 Evaluate all possible actions listed below equally before choosing the most appropriate
 one.
9 Choose the action that best preserves a logical hierarchy, semantic clarity, and
 appropriate abstraction level.
10
11 **Priority Guidance**:
12 1. FIRST consider "go_down" if ANY existing subnode could reasonably contain the new
 topic as a specialization
13 2. THEN consider "make_parent" if multiple existing subnodes could be grouped under a
 new category
14 3. ONLY use "add_sibling" if the topic is FUNDAMENTALLY distinct from all existing
 subnodes at this level
15 4. "discard" should be used for low-quality or redundant topics
16
17 **Critical Rules**:
18 - A node about "Applications of X" should ALWAYS go under X, not as a sibling
19 - Specific methods/tools belong under their parent field (e.g., "PCR" under "Molecular
 Biology")
20 - Avoid creating flat structures
21
22 Possible actions:
23 1) "go_down" - Use this if the topic: {new_topic} is a *more specific* subtype of one of
 the "subnodes" and belongs *within* it.
24 2) "add_sibling" - Use this if the topic: {new_topic} is on the same level of
 abstraction as the existing "subnodes". It should be added *alongside* them as a
 direct child of `{current_path[-1]}`.
25 3) "discard" - Use this if the topic: {new_topic} is irrelevant, redundant, or already
 captured under another topic.
26 4) "make_parent" - Use this when the new topic: {new_topic} or any one of the "subnodes"
 is broader or more abstract than one or more of the subnodes. In that case, make
 the new topic a direct child of `{current_path[-1]}` and move the relevant subnodes
 under it. Return them in `child_nodes`: [...]`.
27
28 ### Example of desired usage for each action:
29 1) "go_down"
30 - "node": must be the name of one of the existing "subnodes"
31 - "explanation": an optional text with reasoning
32 - "child_nodes", "parent_node": not used.
33
34 2) "add_sibling"
35 - "node": {new_topic}
36 - "parent_node": {current_path[-1]}
37 - "explanation": optional
38 - "child_nodes": not used.
39
40 3) "discard"
41 - "node": {new_topic}
42 - "explanation": optional
43 - "parent_node", "child_nodes": not used
44
45 4) "make_parent"
46 - "node": {new_topic} or one of the "subnodes" whichever is more appropriate
47 - "child_nodes": array of the subnodes that must be moved under the new node
48 - "explanation": optional
49 - "parent_node": not used
50
51 Your output must be valid JSON only:
52 {{
53 "action": "go_down"|"add_sibling"|"make_parent"|"discard",
54 "node": "string",
55 "parent_node": "string or null", // only used if action = add_sibling
56 "child_nodes": ["string", ...], // only used if action = make_parent
57 "explanation": "string (optional)"
58 }}
59 No extra text.

```

Figure 11: The detailed prompt used in the execution of our fLMSCI (incremental) baseline. See §4.3 for broader context.

## F Further Details on Collection of Science Papers

This section provides more context on our piles of papers in our experiments from §5.1. **SciPileLarge** is an extension of **SciPile**. For each paper in **SciPile**, we extract five relevant keywords using an LLM (see Fig.12) and query the Semantic Scholar API<sup>3</sup> with these keywords to retrieve additional relevant papers.

We apply three filtering criteria to ensure quality: (a) **Citation Count**: A paper must have a minimum number of citations to be considered reliable. The minimum citation count is calculated using the formula:  $(2 + 3 \times (2025 - \text{publish\_year}))$ . (b) **Abstract Length**: A paper must have an abstract with at least 250 tokens, as measured by the tokenizer of Llama-3.1-8B-Instruct. (c) **Publication Venue**: A paper must be published in a recognized journal or conference. For each keyword, we select up to five papers that meet all criteria. This approach maintains the disciplinary distribution of our seed dataset **SciPile** while expanding our corpus to 10K papers.

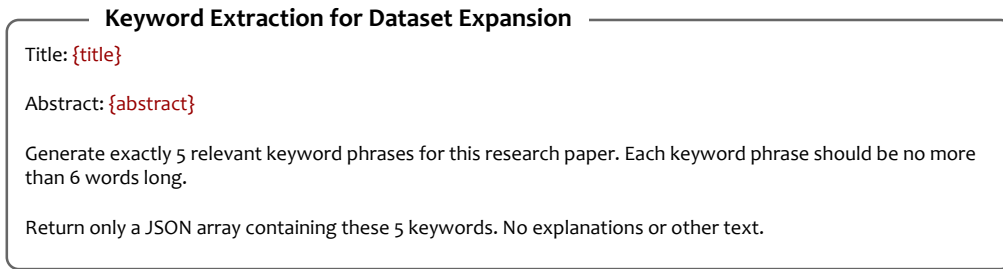


Figure 12: Prompt of Keyword Extraction for Dataset Expansion

## G Hyperparameters of SCYCHIC

We utilize the GPT-4o model (gpt-4o-2024-08-06) to generate all contribution extractions along with detailed rationales explaining the extraction decisions. For **summarizer**, we use We choose Llama-3.3-70B-Instruct for its superiority of following instructions among open-source models, and use gte-Qwen2-7B-instruct as our **embedder**. For clustering algorithm, we apply k-means clustering. The number of clusters for each layer is (6, 40, 276) when conducting experiments on **SciPile** (2K papers), and (6, 40, 276, 1250) when on **SciPileLarge** (10k papers).

<sup>3</sup><https://www.semanticscholar.org/product/api>